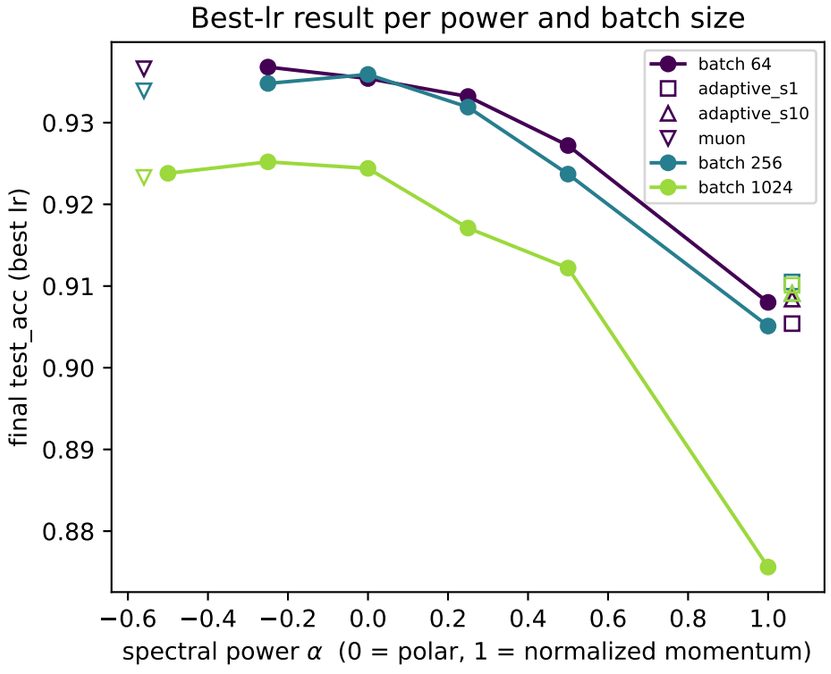
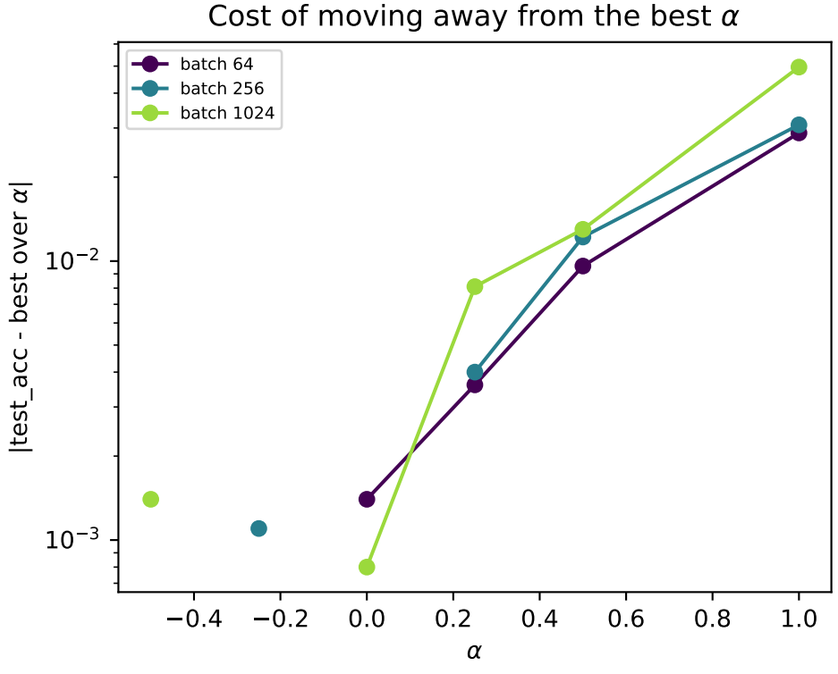
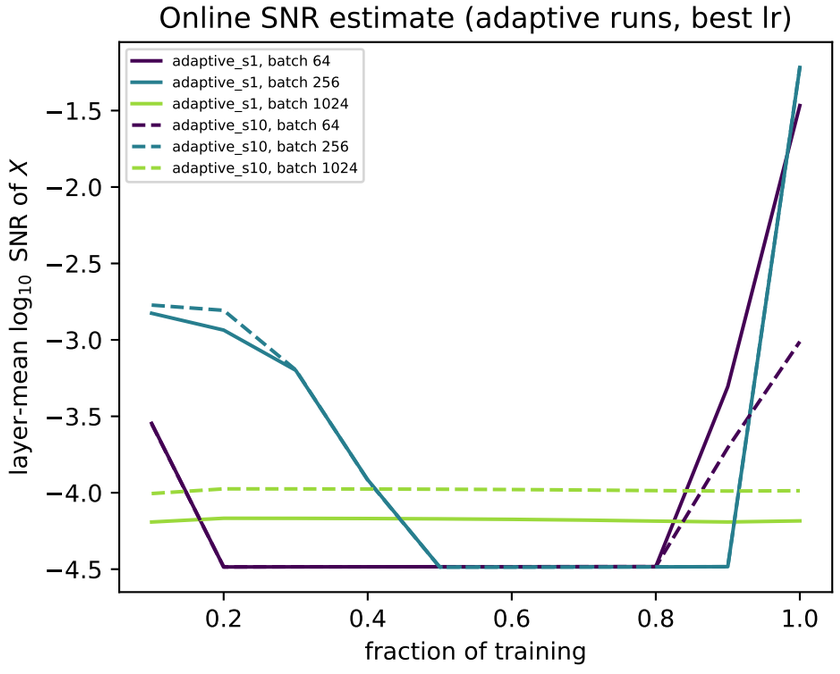

Best-lr result per power and batch size
final test_acc (best lr)
0.93
0.92
0.91
0.90
0.89
0.88
batch 64
adaptive_s1
adaptive_s10
muon
batch 256
batch 1024
−0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8 1.0
spectral power α (0 = polar, 1 = normalized momentum)


Cost of moving away from the best α
|test_acc - best over α|
10^{-2}
10^{-3}
batch 64
batch 256
batch 1024
−0.4 −0.2 0.0 0.2 0.4 0.6 0.8 1.0
α


Online SNR estimate (adaptive runs, best lr)
layer-mean $\log_{10}$ SNR of X
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
−4.5
adaptive_s1, batch 64
adaptive_s1, batch 256
adaptive_s1, batch 1024
adaptive_s10, batch 64
adaptive_s10, batch 256
adaptive_s10, batch 1024
0.2 0.4 0.6 0.8 1.0
fraction of training